

Y4er dotnet-deserialization (notes repository)

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- Published: date not stated
- Original: <https://github.com/Y4er/dotnet-deserialization>
- Preserved from: <https://github.com/Y4er/dotnet-deserialization> (preserved-copy) on 2026-08-04
- Repository commit: c8dac42dfc267d32492e18fe13a3ba5ef717b58c
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README.md

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dotnet deserialization

This series is the author's notes on systematically learning dotnet deserialization from 0 to 1. It covers the official deserialization formatters and the deserialization components of third-party libraries (such as Json.net), interspersed with the usage and principles of ysoserial.net, as well as some dotnet knowledge points.

The author is also a newcomer; if there are mistakes in the wording or explanation of the articles, please do not hesitate to correct me.

All articles are first published on the Xianzhi community

<https://xz.aliyun.com/u/12258>

References

The whole series refers to the following content

1. ysoserial.net
2. docs.microsoft.com
3. <https://github.com/Ivan1ee/NET-Deserialize>

It refers heavily to the articles and Github of [@Ivan1ee](#), as well as Microsoft documentation and some foreign talks and papers, and also the articles of [@pwntester](#).

Contents

1. [dotnet serialize 101](#)

Explains the basics of dotnet serialization and its lifecycle

1. [XmlSerializer](#)

Explains the basics of xmlserializer, the ysoserial.net ObjectDataProvider attack chain and XmlReader.Parse()

1. [BinaryFormatter](#)

Explains the basics of the binary formatter and the TextFormattingRunProperties, DataSet and TypeConfuseDelegate attack chains

1. [Nancy cookie deserialization](#) and the `ToolboxItemContainer` attack chain

1. [SoapFormatter](#)

Explains deserialization vulnerabilities of the soap format stream and the `ActivitySurrogateSelector`, `ActivitySurrogateSelectorFromFile`, `ActivitySurrogateDisableTypeCheck` and `AxHostState` attack chains, plus RCE in Kentico CMS

1. [LosFormatter](#)

Explains `LosFormatter` deserialization, as well as the `ClaimsIdentity`, `WindowsIdentity`, `WindowsClaimsIdentity` and `SessionSecurityToken` attack chains.

1. [ObjectStateFormatter](#)

Explains `ObjectStateFormatter` deserialization as well as the `RolePrincipal` and `WindowsPrincipal` attack chains.

1. [DataContractSerializer](#)

Explains `DataContractSerializer` deserialization, the `SessionViewStateHistoryItem` attack chain, and the abuse of the `DataContractResolver` type resolver.

1. [NetDataContractSerializer](#)

Explains `NetDataContractSerializer` deserialization as well as the `PSObject` attack chain

1. [DataContractJsonSerializer](#)

Explains `DataContractJsonSerializer` deserialization and the `IDataContractSurrogate` interface

1. [JavaScriptSerializer](#)

Explains `JavaScriptSerializer` deserialization

1. [Json.Net](#)

Explains `json.net` deserialization, and gets a deeper understanding through the real-world case breeze CVE-2017-9424.

1. [Fastjson](#)

Explains `fastjson` deserialization vulnerabilities

1. [.NET Remoting](#)

Explains `.net` remoting vulnerabilities

1. [SharePoint CVE-2019-0604](#)

1. [ViewState](#)

Some articles

Below are some articles the author collected while auditing dotnet.

- [CVE-2022-26500 Veeam Backup & Replication RCE](#)
- [Looking at file upload waf bypass from the dotnet source code](#)

- [A few more gadgets for Dotnet deserialization](#)
- [CVE-2021-34992 Orckestra C1 CMS Deserialization RCE](#)
- [Some notes of Microsoft Exchange Deserialization RCE \(CVE-2021-42321\)](#)
- [50 Shades of SolarWinds Orion Deserialization \(Part 1: CVE-2021-35215\)](#)
- https://twitter.com/Y4er_ChaBug/status/1453971672629796865
- [50 Shades of SolarWinds Orion \(Patch Manager\) Deserialization \(Final Part: CVE-2021-35218\)](#)
- [CVE-2022-26503 Veeam Agent for Microsoft Windows LPE](#)
- [Pwning 3CX Phone Management Backends from the Internet](#)
- [HITCON 2023 x DEVCORE Wargame: My todolist Write-up](#)
- [FINDING DESERIALIZATION BUGS IN THE SOLARWIND PLATFORM](#)
- [Exploiting Hardened .NET Deserialization: New Exploitation Ideas and Abuse of Insecure Serialization \(an amazing article\)](#)

About

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